



Proactive AI-Driven Resource Scaling in Cloud Environments

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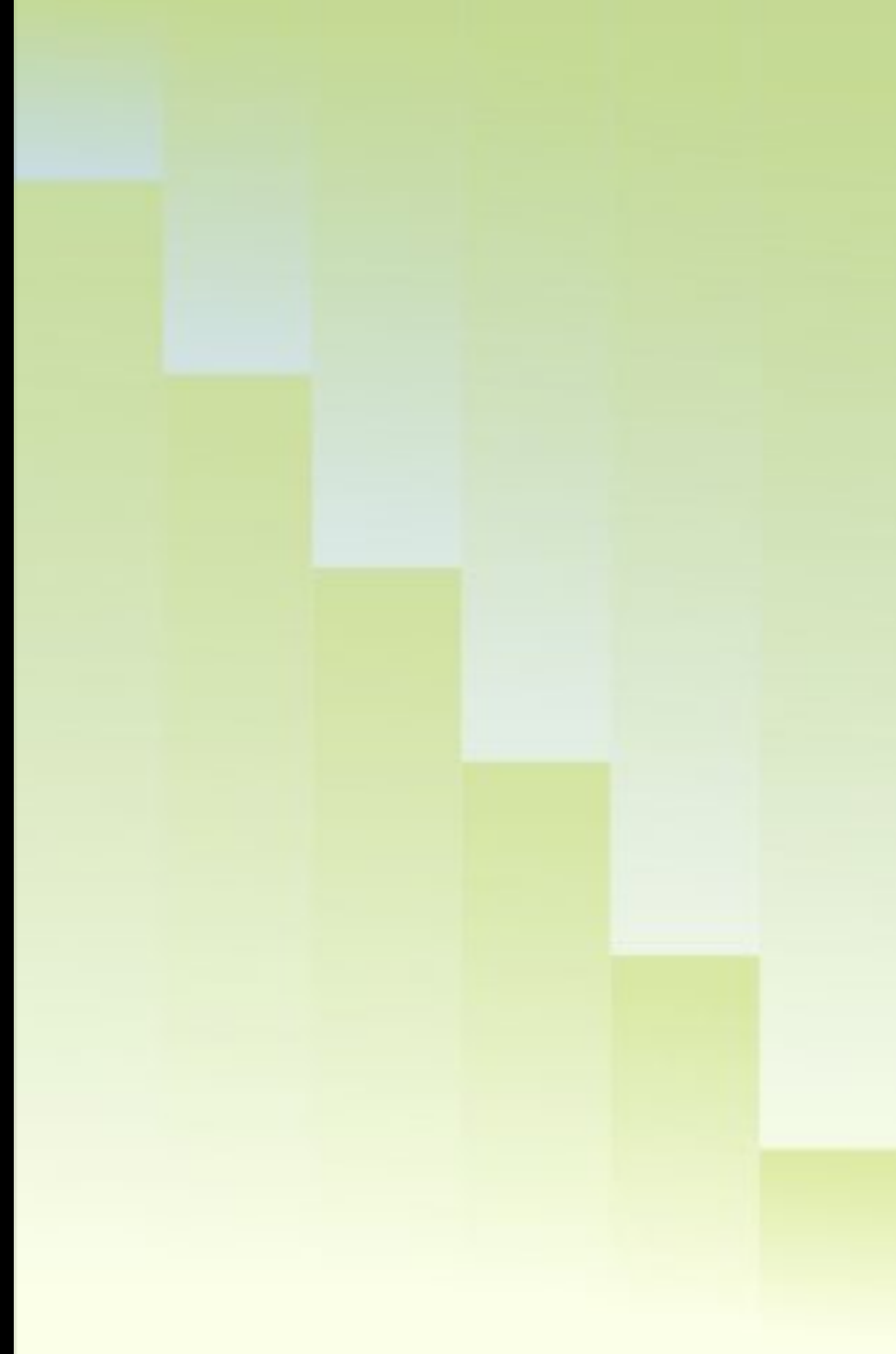
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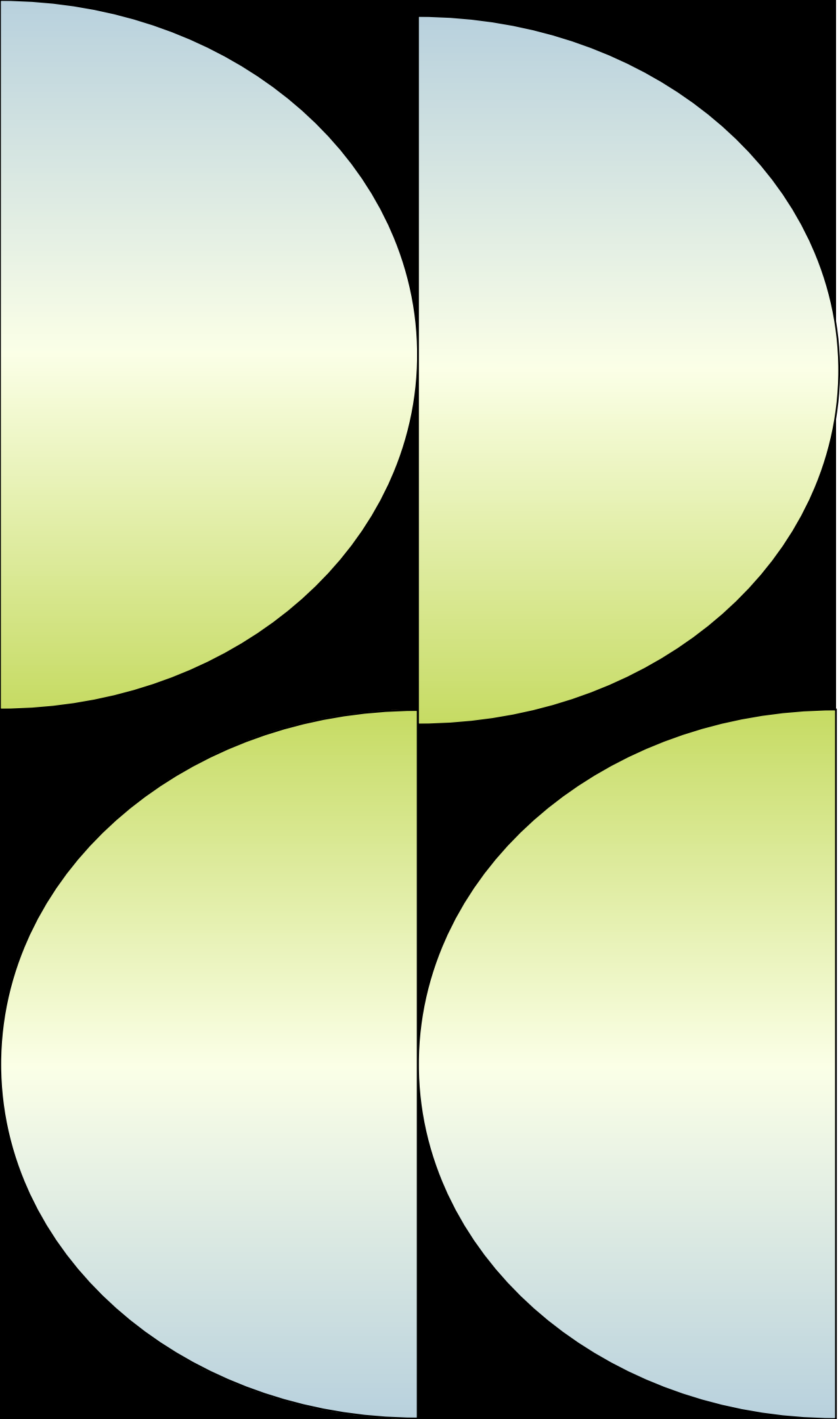
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Motivation: The "Provisioning Wall"

The Trade-off: Balancing cost-effectiveness vs. service reliability (SLA).

Reactive Failure: Rule-based scaling (e.g., >80% CPU) is always "behind" the curve.

The Constraint: 300-second hardware provisioning latency creates a 5-minute "blind spot."

Problem Statement & Goals

Primary Question: "Can an AI-driven policy reduce SLA violations compared to a threshold baseline given fixed provisioning lag?"

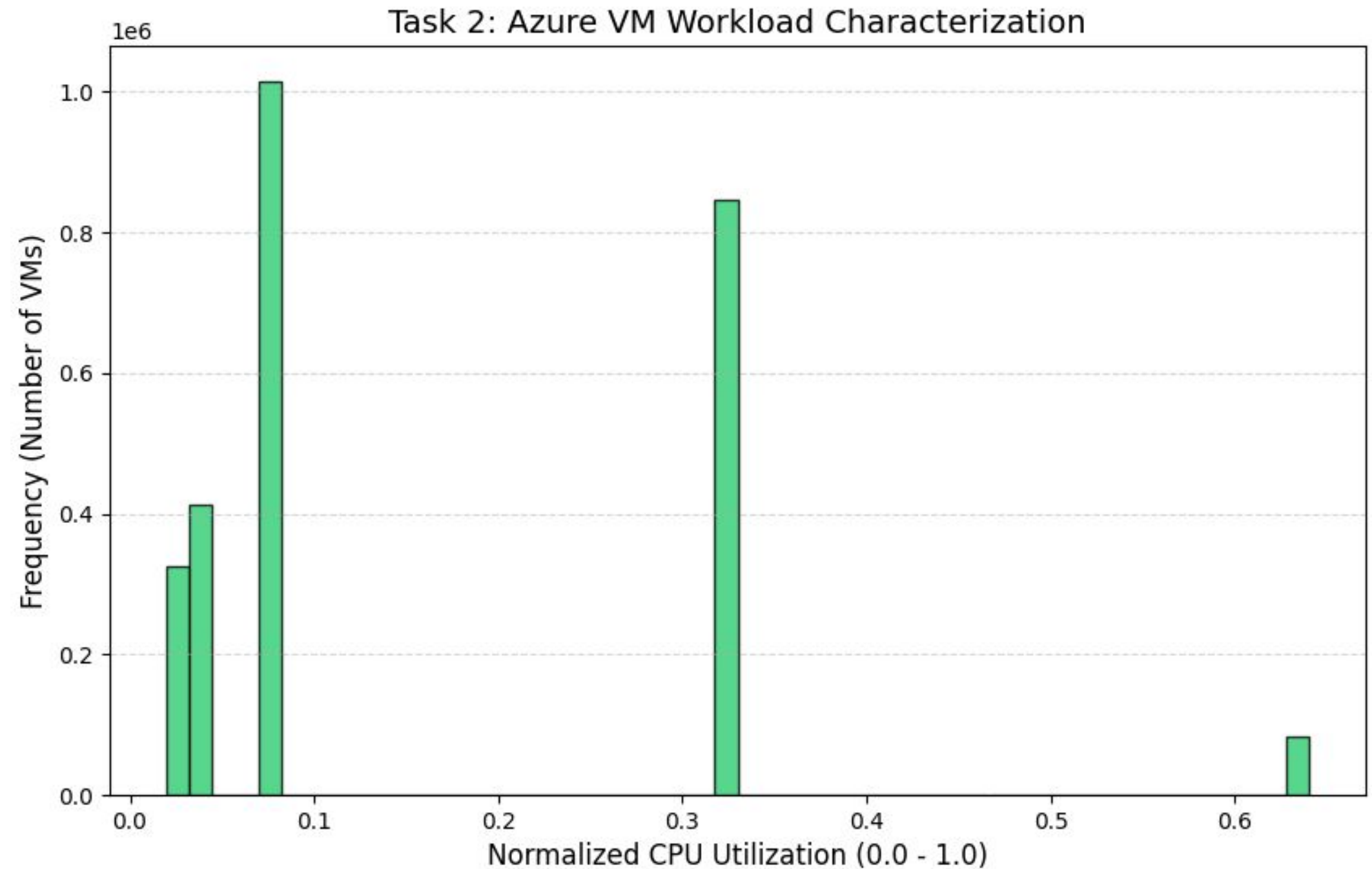
Primary Objective: Evaluate LSTM forecasting on Microsoft Azure public traces.

Dataset & Preprocessing

Source: Microsoft Azure Public Workload Traces (2017/2019).

Nature: Diverse, bursty, multi-tenant subscription-level data.

Preprocessing: Min-Max Normalization; Sliding Window transformation for time-series forecasting.

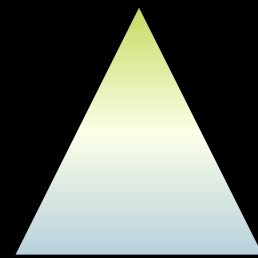


Proposed Solution: Proactive LSTM



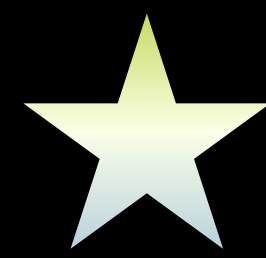
Model

Long Short-Term
Memory (LSTM) for
temporal dependencies



Architecture

Input → Hidden LSTM
Layers → Dropout (10%)
→ Output



Optimization

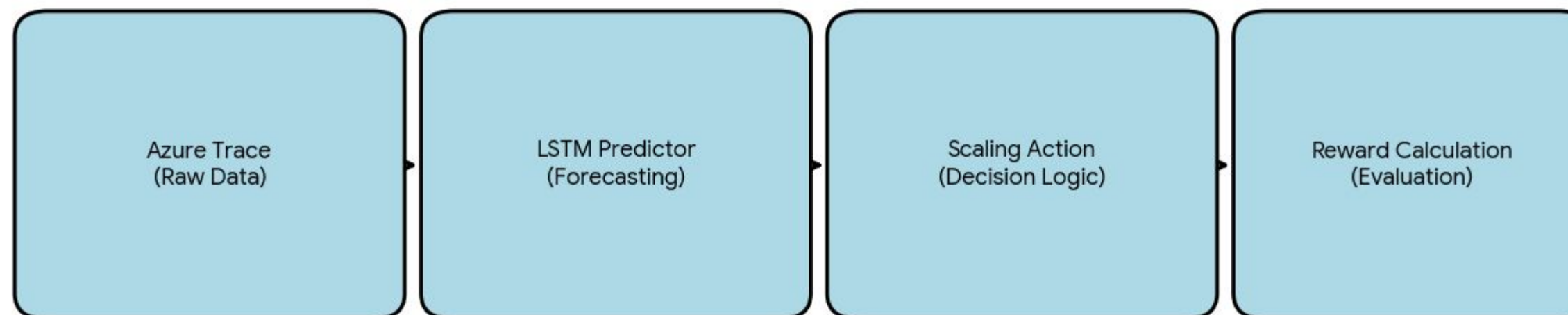
Early Stopping to
prevent overfitting on
specific tenant patterns

Experimental Setup & Assumptions

Assumptions: CPU is the primary health signal; Open-loop simulation (trace-driven).

Environment: Python (PyTorch/TensorFlow).

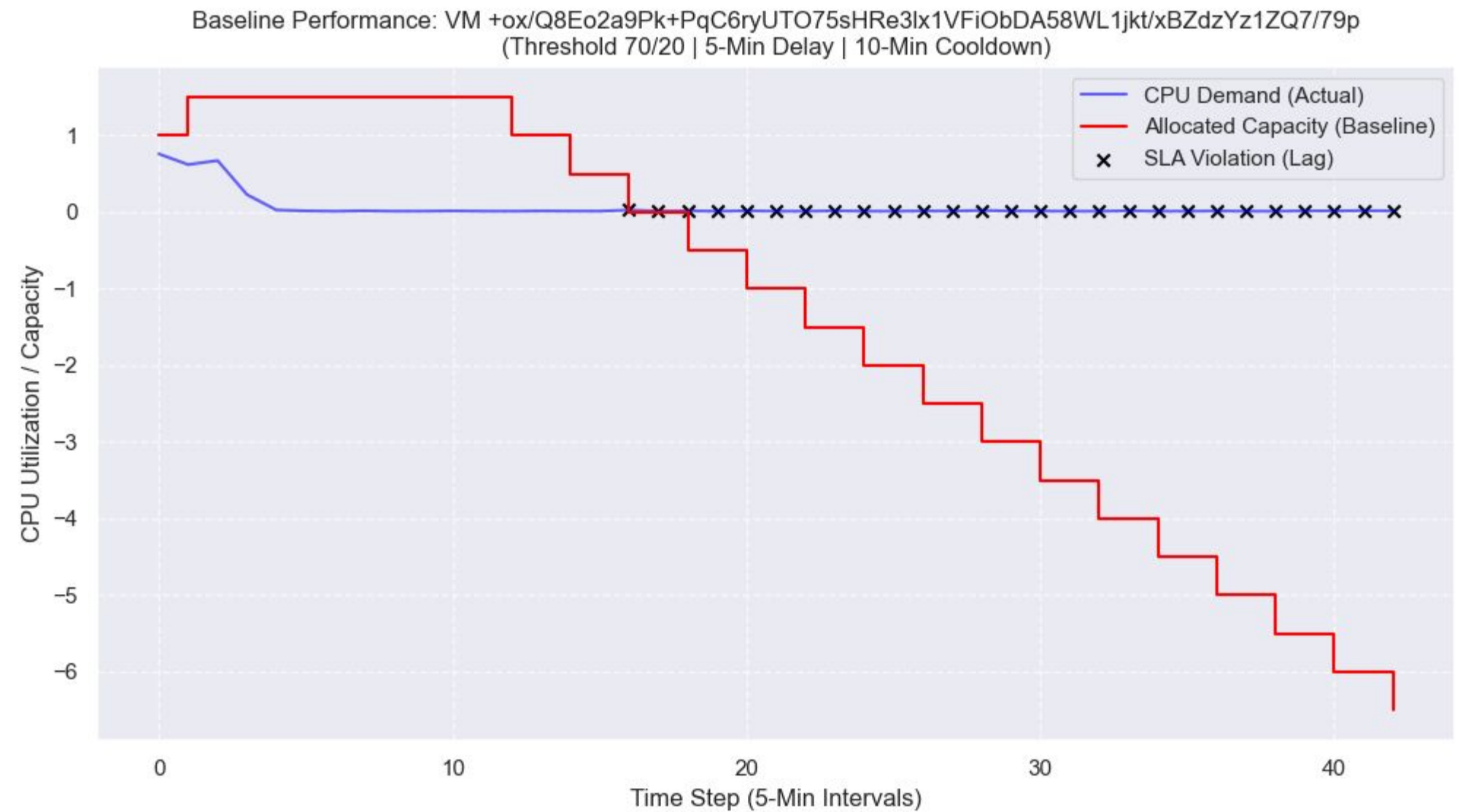
Success Metrics: SLA Violation Rate, Resource Efficiency, Scaling Stability.



Visualizing the Lag (The Baseline)

The Gap: Clearly identifies where the reactive model fails to meet demand.

Impact: This specific gap leads to the SLA violations discussed in the success metrics.

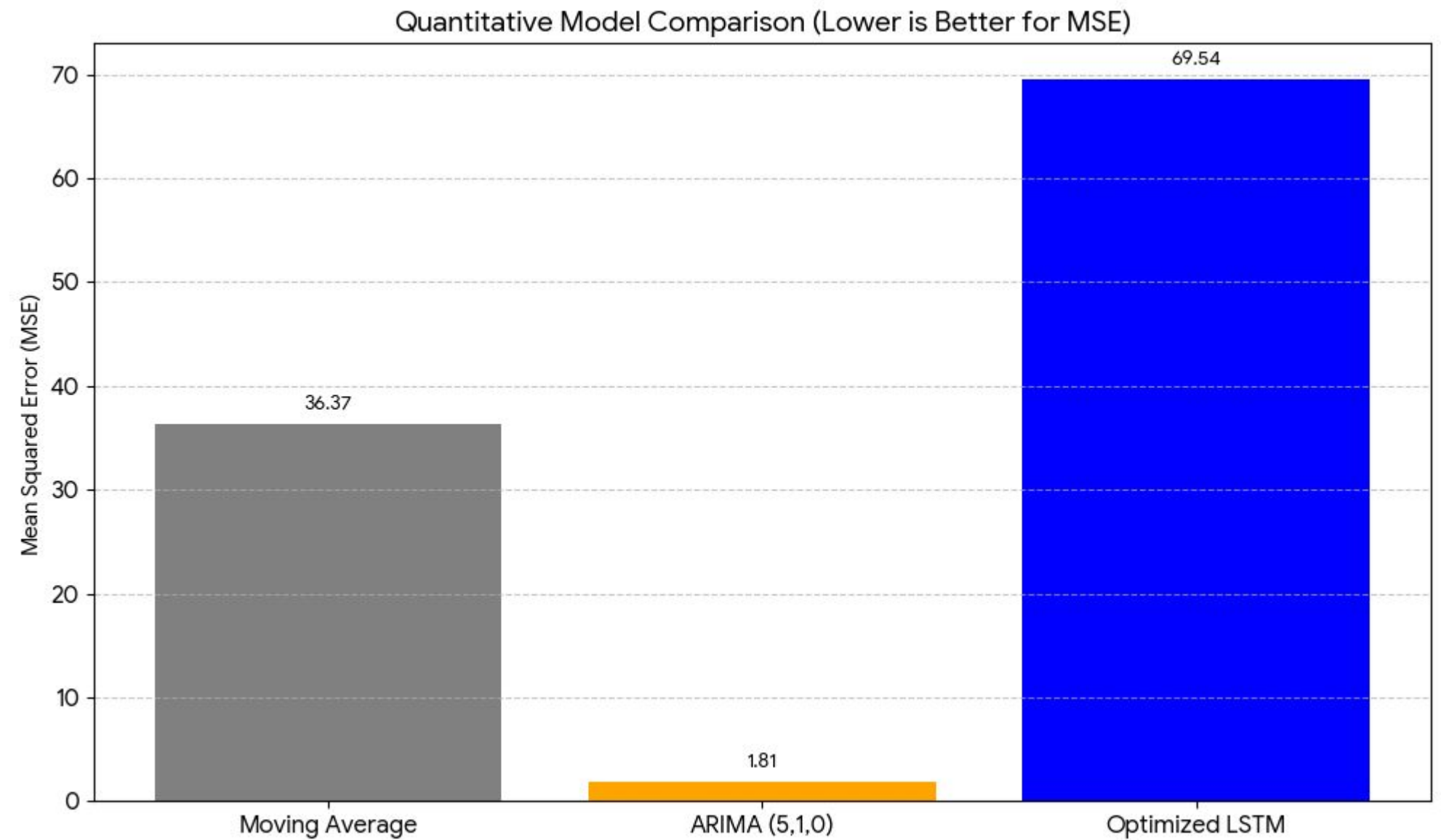


Quantitative Results & Model Comparison

Moving Average (MA): 36.37

ARIMA (5,1,0): 1.81

Proactive LSTM (Optimized): 69.54

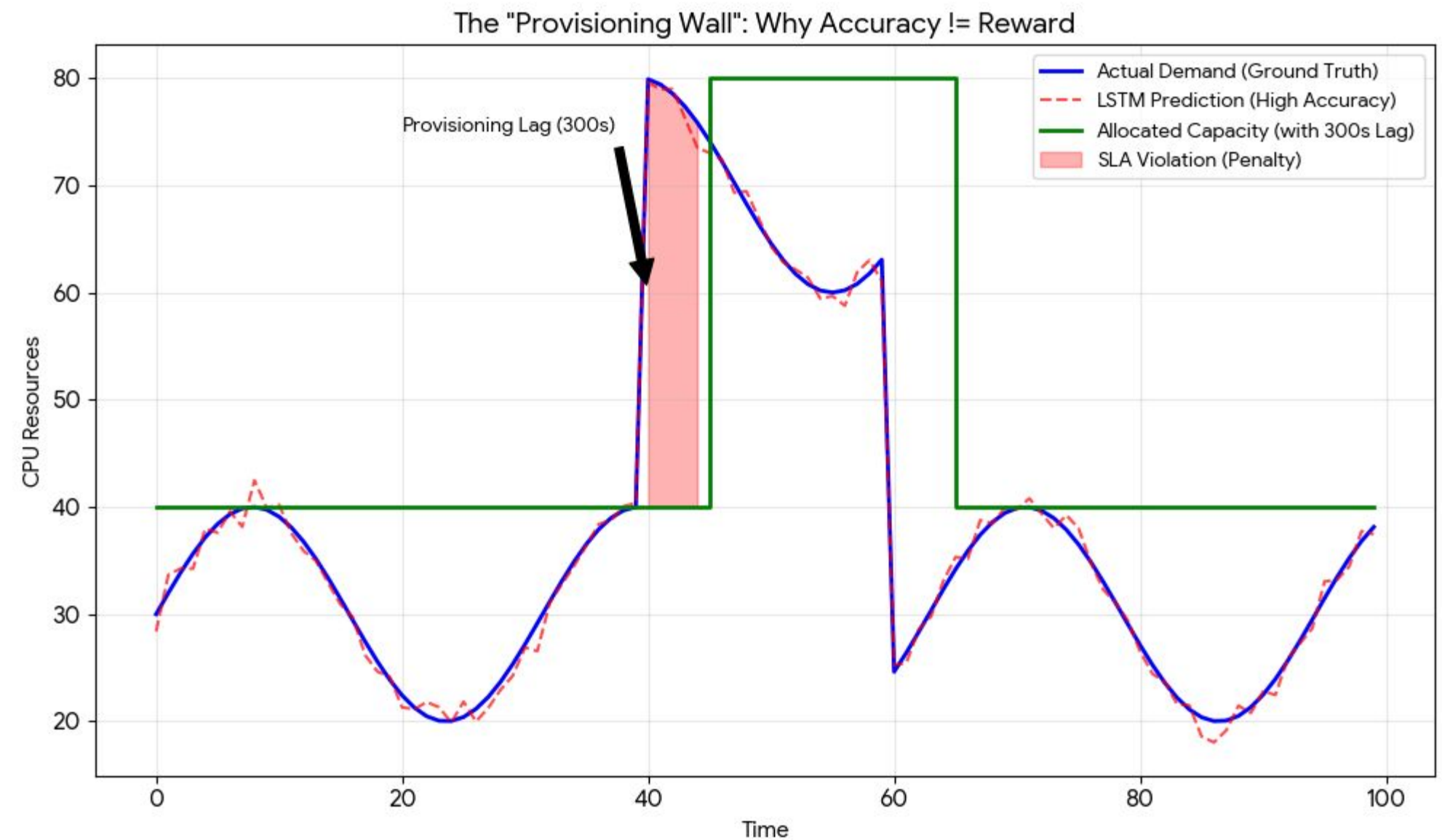


The Hard Truth: System Reward

Key Finding: Average System Reward = -7.8848.

The Analysis: High prediction accuracy does not automatically result in a positive reward.

The Culprit: The 300-second provisioning latency is a physical bottleneck. Even a "perfect" prediction is limited by the time it takes for hardware to respond.

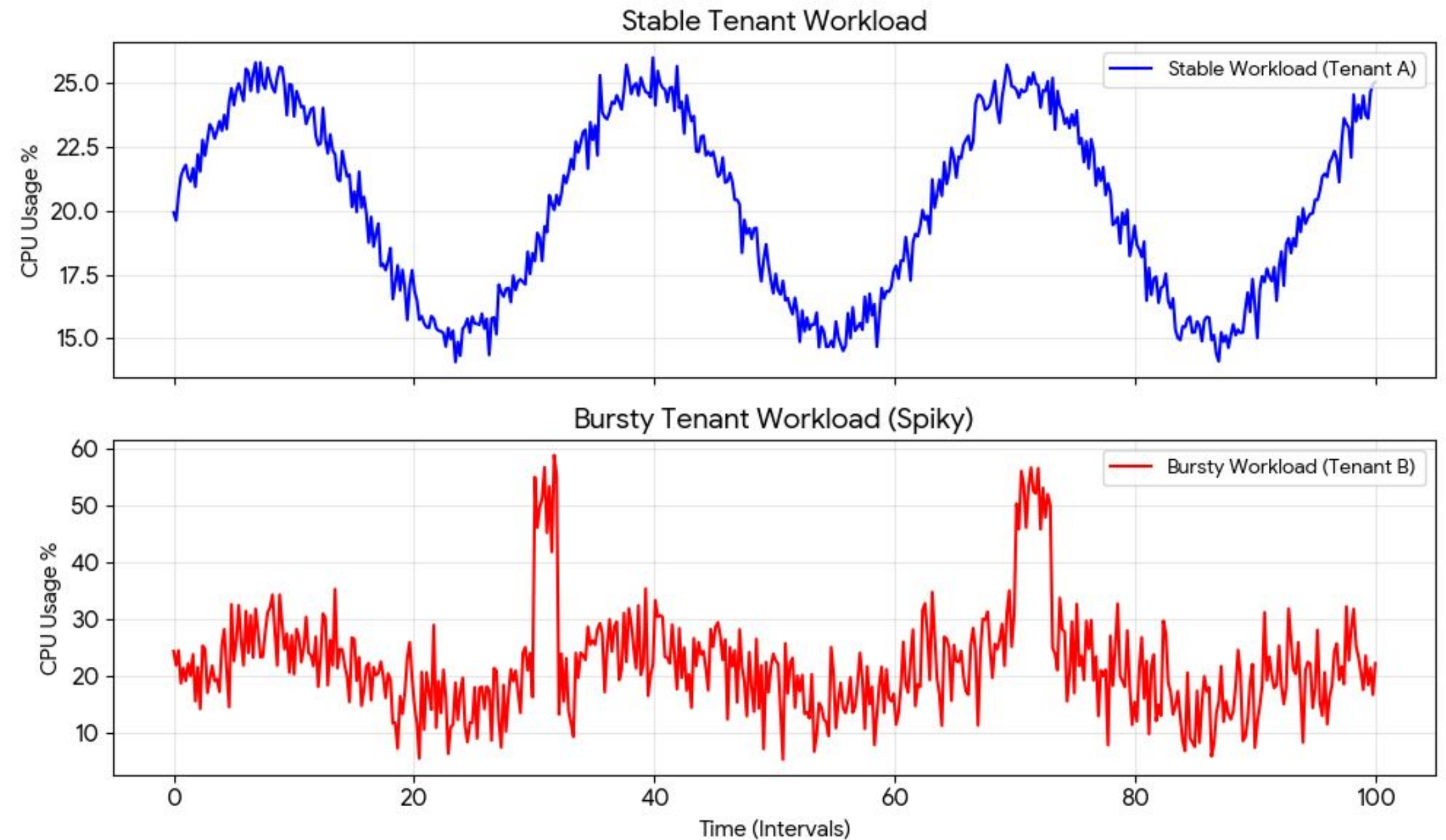


Discussion: The Generalization Gap

The Gap: A model tuned for one subscription (tenant) often fails when applied to others due to extreme workload variance.

Data Evidence: Refer back to the Azure Trace diversity (your histogram).

Insight: Moving from "Localized" to "Global" models is the primary hurdle for cloud-native AI.



Conclusion & Future Directions

Summary: Forecasting is a "strategic enabler," but policy optimization is the next frontier.

Future Work: Reinforcement Learning (RL): To optimize the policy, not just the prediction.

Reproducibility: GitHub:

<https://github.com/jayden-graneta/Al-Driven-Cloud-Optimization>